

21. What is a String in Python?

Answer:

A **string** is a sequence of characters enclosed in single quotes (' '), double quotes (" "), or triple quotes (''' ' or "" "" """).

Strings are **immutable**, meaning they cannot be changed after they are created.

Example:

```
name = "Bhagya"
```

```
print(name)
```

Output:

Bhagya

Interview Tip:

Strings are one of the most commonly used data types in Python and are frequently asked about in interviews.

22. What is String Indexing?

Answer:

Indexing is used to access individual characters in a string.

- Positive indexing starts from 0.
- Negative indexing starts from -1.

Example:

```
text = "Python"
```

```
print(text[0]) # P
```

```
print(text[3]) # h
```

```
print(text[-1]) # n
```

Output:

P

h

n

Interview Tip:

Python uses zero-based indexing.

23. What is String Slicing?**Answer:**

Slicing extracts a part of a string.

Syntax:

```
string[start:end:step]
```

- start → Starting index
- end → Ending index (not included)
- step → Number of characters to skip

Example:

```
text = "Python"
```

```
print(text[1:4])
```

```
print(text[:3])
```

```
print(text[3:])
```

Output:

```
yth
```

```
Pyt
```

```
hon
```

Interview Tip:

The ending index is **excluded** in slicing.

24. What are some commonly used String Methods?**Answer:**

Python provides many built-in string methods.

Method Description

upper() Converts to uppercase
lower() Converts to lowercase
strip() Removes extra spaces
replace() Replaces text
split() Splits a string into a list
find() Finds the position of a substring

Example:

```
text = "python interview"
```

```
print(text.upper())  
print(text.replace("python", "Java"))
```

Output:

```
PYTHON INTERVIEW  
Java interview
```

Interview Tip:

String methods are very common in coding rounds.

25. What is the difference between append() and extend() in Python Lists?

Answer:

Both methods add elements to a list, but they work differently.

- **append()** adds the entire object as a **single element**.
- **extend()** adds each element of an iterable **individually**.

Example:

```
a = [1, 2]  
a.append([3, 4])  
print(a)
```

```
b = [1, 2]  
b.extend([3, 4])  
print(b)
```

Output:

```
[1, 2, [3, 4]]
```

```
[1, 2, 3, 4]
```

Interview Tip:

This is one of the most frequently asked Python interview questions for freshers.